Measuring Prosody Diversity in Zero-Shot TTS: A New Metric, Benchmark, and Exploration

Yifan Yang[1⋆], Bing Han[1⋆], Hui Wang[3], Long Zhou[2†], Wei Wang[1], Mingyu Cui[2], Xu Tan[2], Xie Chen[1,4‡]
[1]X-LANCE Lab, Shanghai Jiao Tong University [2]Tencent Hunyuan [3]Nankai University [4]Shanghai Innovation Institute
⋆Equal Contribution †Project Leader ‡Corresponding Author

Motivation & Problem

Prosody diversity is essential for naturalness and expressiveness in zero-shot TTS, yet remains underexplored.

Limitations of existing metrics:

- Log F_0 RMSE: captures pitch only, ignores rhythm & intensity & duration, requires costly DTW, correlates weakly with human perception
- MCD correlates relatively better with human ratings, but also relies on costly DTW
- No reliable automatic metric for prosody diversity that aligns with human judgment

Recent advances:

- SSL speech representations (HuBERT, WavLM) effectively encode prosodic information
- NLP metrics adapted to discrete speech (SpeechTokenDistance) remain preliminary

Gap: We need an objective metric for prosody diversity that is accurate, efficient, and aligns with human perception.

Research Questions:

- Can we design a metric that better correlates with human prosody diversity ratings?
- Which TTS architectures produce more diverse prosody?
- Do RL-based alignment methods reduce prosody diversity?
- Can large audio language models (LALMs) reliably evaluate prosodic variation?

ProsodyEval Dataset

Human-annotated dataset for prosody diversity assessment in zero-shot TTS.

Property	Value
Speech samples	1,000
Human ratings	2,000
TTS systems	7
Raters	20 graduate students (TTS research experience)

7 TTS Systems:

- AR: XTTS-v2, CosyVoice, CosyVoice 2
- NAR flow matching: E2 TTS, F5-TTS, ZipVoice
- NAR masked generative modeling: MaskGCT

Protocol: Each system generates 5 samples per input with random seeds 0–4.

- **PMOS:** Raters score pairwise prosodic difference (1–5) across all 10 pairs within a group
- 1 = nearly identical prosody 5 = clearly distinct speaking styles
- Acoustic baselines: log F_0 RMSE and MCD computed with FastDTW

DS-WED Metric

DS-WED = Discretized Speech Weighted Edit Distance

Pipeline:

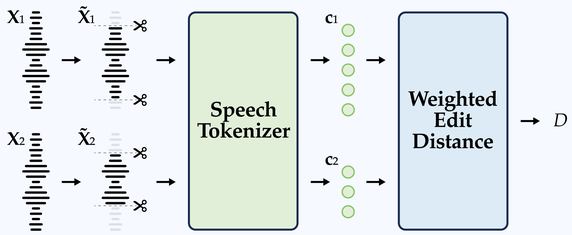
1. Trim leading and trailing non-speech segments from audio using Silero VAD

2. SSL encoder (HuBERT/WavLM) + k-means → token sequences → Unicode Chinese character sequences $\mathbf{c}_1, \mathbf{c}_2$

3. Weighted Levenshtein distance:

$$\text{DS-WED}(\mathbf{c}_1, \mathbf{c}_2) = \min_{\pi \in \mathcal{A}(\mathbf{c}_1, \mathbf{c}_2)} \sum_{(i,j,o) \in \pi} w_o \, c_o(c_{1,i}, c_{2,j})$$

where $o \in \{\text{sub}, \text{ins}, \text{del}\}$, $w_o = 1$ in all experiments; $w_{\text{sub}} = 1.2$ better aligns with human judgment (on ProsodyEval)



Why semantic tokens?

- HuBERT/WavLM tokens encode prosodic information effectively
- GPU-friendly: no costly DTW required
- Acoustic tokens (EnCodec) not used: retain low-level signal details irrelevant to prosody

Default config: HuBERT-base, layer 8, 50 clusters, $c_{\text{sub}} = c_{\text{ins}} = c_{\text{del}} = 1$

Correlation with Human Judgments

DS-WED achieves the highest correlation with PMOS ($\bar{r} = 0.77$), substantially outperforming acoustic metrics.

- Correlations: average Pearson $\bar{r}$, aggregated via Fisher's Z transformation
- Brackets: 95% confidence intervals (Fisher space, back-transformed)
- Darker green = stronger correlation
- all $p < 0.001$

	PMOS	log F_0 RMSE	MCD	**DS-WED**
PMOS	-	$0.30_{[0.19,\,0.40]}$	$0.66_{[0.58,\,0.73]}$	$\mathbf{0.77}_{[0.73,\,0.81]}$
log F_0 RMSE	$0.30_{[0.19,\,0.40]}$	-	$0.35_{[0.25,\,0.44]}$	$0.36_{[0.26,\,0.45]}$
MCD	$0.66_{[0.58,\,0.73]}$	$0.35_{[0.25,\,0.44]}$	-	$0.82_{[0.74,\,0.87]}$
DS-WED	$\mathbf{0.77}_{[0.73,\,0.81]}$	$0.36_{[0.26,\,0.45]}$	$0.82_{[0.74,\,0.87]}$	-

Efficiency (RTF on A100 80G):

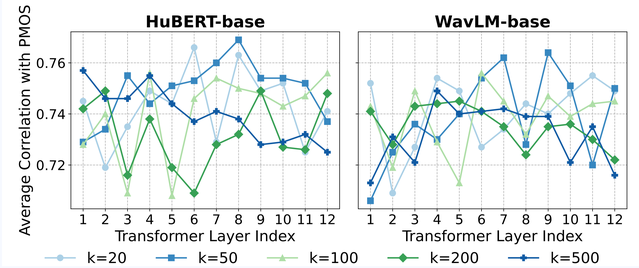
Metric	RTF ↓	GPU-friendly
Log F_0 RMSE	0.549	No (CPU-bound)
MCD	0.203	No (CPU-bound)
DS-WED	**0.110**	**Yes**

DS-WED is **5× faster** than log F_0 RMSE and **2× faster** than MCD, with GPU acceleration for large-scale evaluation.

Ablation Study of DS-WED

DS-WED is **robust** across SSL models, encoder layers, and cluster sizes (correlations consistently within 0.7-0.8).

- Middle layers (6–9) achieve stronger correlations, encoding richer prosodic information
- Relative Smaller cluster sizes (50) perform best; larger sizes reduce peak correlations, making the edit-distance calculation overly sensitive and misaligned with human perceptual sensitivity to prosody
- WavLM-base more stable, HuBERT-base larger variance
- HuBERT-large and WavLM-large yield slightly higher correlations, but at much greater cost
- HuBERT-base layer 8 with 50 clusters achieves the highest peak $r = 0.77$



Prosody Diversity Benchmark

First systematic benchmark across current state-of-the-art open-source TTS systems.
Evaluated on **LibriSpeech test-clean** and **Seed-TTS test-en**.

Type	System	LibriSpeech *test-clean*						Seed-TTS *test-en*					
		log F_0 RMSE		MCD		DS-WED		log F_0 RMSE		MCD		DS-WED	
		Avg.↑	*Borda*↑	*Avg.*↑	*Borda*↑	*Avg.*↑	*Borda*↑	*Avg.*↑	*Borda*↑	*Avg.*↑	*Borda*↑	*Avg.*↑	*Borda*↑
AR	XTTS-v2	**0.31**	**5.14**	4.18	4.70	127.84	4.89	**0.28**	**5.73**	4.12	4.16	**93.15**	5.50
	CosyVoice	0.27	3.46	4.08	4.41	120.59	4.59	0.22	3.49	4.29	4.85	75.74	4.85
	CosyVoice 2	0.30	4.56	4.07	4.47	134.34	5.38	0.24	4.45	4.12	4.35	88.04	**5.78**
NAR flow	E2 TTS	0.27	3.26	3.40	1.87	84.91	2.11	0.20	2.78	3.33	1.57	52.35	2.18
	F5-TTS	0.26	2.98	3.48	2.19	79.59	1.50	0.21	3.23	3.49	2.28	49.00	1.51
	ZipVoice	0.29	4.55	3.91	3.85	114.52	3.93	0.22	3.55	3.99	4.09	58.56	2.88
NAR MGM	MaskGCT	0.28	4.04	**4.76**	**6.51**	**139.75**	**5.61**	0.24	4.78	**5.13**	**6.70**	80.36	5.30

- Modeling paradigm rather than data scale is the dominant factor
- AR and NAR masked generative models consistently outperform NAR flow-matching
- MaskGCT achieves highest DS-WED on LibriSpeech; CosyVoice 2 achieves highest DS-WED on Seed-TTS

Finding 1: Duration Variation Shapes Prosody

Q: Does duration variation shape prosody diversity in NAR systems?
Duration perturbation (DP) with factors {0.8, 0.9, 1.0, 1.1, 1.2}:

System	LibriSpeech *test-clean*	Seed-TTS *test-en*
	DS-WED *Avg.*↑	DS-WED *Avg.*↑
F5-TTS	79.59	49.00
F5-TTS w/ DP	$100.88_{+26.7\%}$	$62.95_{+28.5\%}$
MaskGCT	139.75	80.36
MaskGCT w/ DP	$159.10_{+13.8\%}$	$92.71_{+15.4\%}$

DP consistently increases prosody diversity for two NAR systems: duration variation during inference is a key factor shaping prosody.

F5-TTS gains +28.5% with duration perturbation, yet still lags behind AR/MGM without DP: **flow-matching with implicit alignment has inherent architectural limitations**.

Finding 2: RL Reduces Prosody Diversity

Q: Does reinforcement learning reduce prosody diversity?
Systems aligned via **DPO** (Direct Preference Optimization) on INTP dataset for intelligibility:

System	LibriSpeech *test-clean*	Seed-TTS *test-en*
	DS-WED *Avg.*↑	DS-WED *Avg.*↑
CosyVoice 2	134.34	88.04
CosyVoice 2 w/ DPO	$109.09_{-18.8\%}$	$71.64_{-18.6\%}$
MaskGCT	139.75	80.36
MaskGCT w/ DPO	$135.75_{-2.9\%}$	$77.80_{-3.2\%}$

DPO consistently reduces prosodic diversity: RL prunes variation while amplifying reward-aligned (intelligibility) behaviors, forming a fundamental trade-off between diversity and task-specific optimization.

Finding 3: Current LALMs Are Unreliable Evaluators

Q: Can large audio language models (LALMs) evaluate prosodic differences?
Gemini 2.5 Pro prompted to rate prosodic difference (1–5) within groups of 5 samples:

	PMOS	log F_0 RMSE	MCD	DS-WED
Gemini 2.5 Pro	$0.27^*_{[0.16,\,0.38]}$	$0.10_{[-0.08,\,0.27]}$	$0.16^*_{[0.01,\,0.30]}$	$0.22^*_{[0.05,\,0.38]}$

- High prompt sensitivity: results vary substantially with minor prompt changes
- LALMs lack a consistent internal representation of prosodic variation

Gemini 2.5 Pro is not a reliable evaluator of prosodic variation. Automatic metrics like DS-WED provide more consistent and reproducible assessments.

Conclusions

1. DS-WED Metric: Reliable prosody diversity metric via weighted edit distance on semantic tokens that outperforms acoustic metrics ($\bar{r} = 0.77$ vs 0.66/0.30), GPU-friendly (RTF=0.110), robust across models/layers/clusters.

2. First systematic benchmark of prosody diversity across state-of-the-art open-source TTS systems.

3. Key insights:

- AR > NAR flow-matching: explicit duration modeling matters for diversity; MGM competitive with AR
- DPO trades prosody diversity for intelligibility: a fundamental RL trade-off
- Current LALMs (Gemini 2.5 Pro) are unreliable for prosody evaluation

4. Limitation: validated on English only; cross-lingual applicability left for future work.

Recent Work [ACL 2026 Main]
on Human-Aligned Automatic Metrics for Speech Style Similarity
Yifan Yang, Bing Han, et al. Towards Fine-Grained and Multi-Granular Contrastive Language-Speech Pre-training
arxiv 2601.03065

Audio Samples
prosodyeval.github.io

Code & Models
github.com/yfyeung/DS-WED

LinkedIn
Yifan Yang

WeChat
yifanyeung